KAIZEN MATHS

Lesson starter

Midpoint and Length of Line Segments

Length of a Line Segment

Name _____

S1

Find the length of AB where A (3, 2) and B (6, 2) .

S2

Find the length of AB where A (2, 0) and B (4, 5) . Give an exact answer.

S3

Find the length of AB where A (3, -2) and B (6, -2) .

S4

Find the length of AB where A (-2, -5) and B (4, -2) . Give an exact answer.

Show your working

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GEO-3384-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Length of a Line Segment

Name

E1

Find the length of AB where A (-5, 2) and B (5, 2) .

E2

Find the length of AB where A (0, -3) and B (5, -1) . Give an exact answer.

E3

Find the length of AB where A (-2, -3) and B (4, -3) .

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Length of a Line Segment

Name

E1

Find the length of AB where A (-5, 2) and B (5, 2) .

E2

Find the length of AB where A (0, -3) and B (5, -1) . Give an exact answer.

E3

Find the length of AB where A (-2, -3) and B (4, -3) .

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Show your working

KAIZEN MATHS

Exit ticket

Midpoint and Length of Line Segments

Length of a Line Segment

Name

E1

Find the length of AB where A (-5, 2) and B (5, 2) .

E2

Find the length of AB where A (0, -3) and B (5, -1) . Give an exact answer.

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Find the length of AB where A (-2, -3) and B (4, -3) .

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Show your working

Teacher answers & working

Midpoint and Length of Line Segments · Length of a Line Segment

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S1

Find the length of AB where A (3, 2) and B (6, 2) .

Answer:

3 units

Working:

The y-coordinates are the same, so this is a horizontal distance.

Length = |6 - (3)|

Length = 3

Therefore AB = 3 units.

S2

Find the length of AB where A (2, 0) and B (4, 5) . Give an exact answer.

Answer:

√29 units

Working:

Use the distance formula. This is Pythagoras applied to the horizontal and vertical changes.

$AB = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$AB = \sqrt{((4 - (2))^2 + (5 - (0))^2)}$

$AB = \sqrt{(2^2 + 5^2)}$

$AB = \sqrt{29}$

Therefore AB = √29 units.

S3

Find the length of AB where A (3, -2) and B (6, -2) .

Answer:

3 units

Working:

The y-coordinates are the same, so this is a horizontal distance.

Length = |6 - (3)|

Length = 3

Therefore AB = 3 units.

Teacher answers & working

Midpoint and Length of Line Segments · Length of a Line Segment

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S4

Find the length of AB where A (-2, -5) and B (4, -2) . Give an exact answer.

Answer:

√45 units

Working:

Use the distance formula. This is Pythagoras applied to the horizontal and vertical changes.

$AB = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$AB = \sqrt{((4 - (-2))^2 + (-2 - (-5))^2)}$

$AB = \sqrt{(6^2 + 3^2)}$

$AB = \sqrt{45}$

Therefore AB = √45 units.

E1

Find the length of AB where A (-5, 2) and B (5, 2) .

Answer:

10 units

Working:

The y-coordinates are the same, so this is a horizontal distance.

Length = |5 - (-5)|

Length = 10

Therefore AB = 10 units.

E2

Find the length of AB where A (0, -3) and B (5, -1) . Give an exact answer.

Answer:

√29 units

Working:

Use the distance formula. This is Pythagoras applied to the horizontal and vertical changes.

$AB = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$AB = \sqrt{((5 - (0))^2 + (-1 - (-3))^2)}$

$AB = \sqrt{(5^2 + 2^2)}$

$AB = \sqrt{29}$

Therefore AB = √29 units.

Teacher answers & working

Midpoint and Length of Line Segments · Length of a Line Segment

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E3

Find the length of AB where A (-2, -3) and B (4, -3) .

Answer:

6 units

Working:

The y-coordinates are the same, so this is a horizontal distance.

Length = |4 - (-2)|

Length = 6

Therefore AB = 6 units.